# A Catalog of Ghosts: the 120 Pilot Candidate Counterexamples and Sturmian Realizability Data

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**Abstract**

This catalog accompanies the Shadow Frontier project (successor to *Finite Spectral Shadows for the Collatz Valuation Cocycle*). It lists the 120 candidate-counterexample valuation words of pilot run 001 (deterministic regeneration, seed 1): words of length $M = 240$ over the alphabet $\{1, 2, 4\}$, calibrated to mean $\log_2 3$ within $2/M$ and carrying a non-coboundary order-3 shadow bias, sampled from the law $p_1 = 0.567$, $p_2 = 0.357$, $p_4 = 0.076$. Every one of the 120 is a *mixed-adic ghost*: its 2-adic realizability class modulo $2^{B_M+1}$ has least positive representative of nearly full bit length, not stabilizing between prefix lengths $M/2$ and $M$ — the signature that no positive integer realizes the word. We also list ten Sturmian ghosts of the critical slope $\alpha = 2 - \log_2 3$ (all Sturmian words are now *provably* non-realizable, by the repetition-rigidity theorem); their data is shown for comparison. The objects below are points of $\mathbb{Z}_2$; the Collatz conjecture asserts that classes like these never meet $\mathbb{Z}^+$.

## 1 Conventions

For a valuation word $w = (b_0, \ldots, b_{M-1})$, $B_N = \sum_{n<N} b_n$; the set of odd $x$ whose first $N$ valuations equal the prefix is a single residue class modulo $2^{B_N+1}$; "rep bits" is the bit length of its least positive representative $r_N$, "mod bits" is $B_N + 1$, and the fill ratio is their quotient. A positive realization would force $r_N$ to stabilize at $\lceil \log_2 x_0 \rceil$ once $2^{B_N} > x_0$; a ghost rides the diagonal $r_N \sim 2^{B_N}$. The bias is $|D_3| = \left| \frac{1}{M} \sum_n \zeta_3^{-b_n} - \mathbb{E}_{\text{geom}}[\zeta_3^{-b}] \right|$, the order-3 non-coboundary two-point shadow.

## 2 Translating between a word and the integer it represents

### Explanation

A word does not represent a single integer but a single *residue class*: the set of odd $x$ whose first $M$ valuations equal $w = (b_0, \ldots, b_{M-1})$ is exactly one arithmetic progression

$$x \equiv r \pmod{2^{B_M+1}}, \qquad B_M = \textstyle\sum_{n<M} b_n,$$

and "the integer" tabulated in this catalog is its least positive representative $r$. Every member $r + k\,2^{B_M+1}$ ($k \geq 0$) produces the same first $M$ letters; the word sees only $x \bmod 2^{B_M+1}$. The converse direction is trivial: iterate $x \mapsto (3x+1)/2^{v_2(3x+1)}$ and read the exponents.

### Formula

Along the prefix the orbit point has the exact affine form

$$x_n = \frac{3^n x_0 + A_n}{2^{B_n}}, \qquad A_n = \sum_{j<n} 3^{n-1-j}\, 2^{B_j}, \qquad B_n = \sum_{j<n} b_j.$$

The letter condition $v_2(3x_n+1)=b_n$ is the congruence

$$3^{n+1}x_0 + A_{n+1} \equiv 2^{B_{n+1}} \pmod{2^{B_{n+1}+1}} \quad\iff\quad x_0 \equiv 3^{-(n+1)}\bigl(2^{B_{n+1}} - A_{n+1}\bigr) \pmod{2^{B_{n+1}+1}},$$

(3 is a unit in $\mathbb{Z}_2$), so each letter appends $b_n$ new binary digits of $x_0$: a Hensel-style lift. After $M$ letters the class modulus is $2^{B_M+1}$ and $r$ is the lift's value reduced to $[1, 2^{B_M+1})$.

## Python

The lift as implemented in `search/frontier.py` (`realize`); $r$ is built letter by letter:

```
def realize(bs):                          # word -> (r, B+1)
    r, Mb = 1, 1                          # class rep and its bit-modulus
    A = 0; B = 0; P3 = 1                  # running A_n, B_n, 3^n
    for b in bs:
        need = B + b; mod = 1 << (need + 1)
        c    = (3*A + (1 << B)) % mod          # A_{n+1} mod 2^{B+b+1}
        x0   = (pow(3*P3, -1, mod)
                * (((1 << need) - c) % mod)) % mod
        if need + 1 > Mb:
            r = x0; Mb = need + 1
        A = 3*A + (1 << B); P3 *= 3; B += b
    return r, Mb

def word_of(x, M):                        # integer -> word
    out = []
    for _ in range(M):
        t = 3*x + 1; b = (t & -t).bit_length() - 1
        out.append(b); x = t >> b
    return out
```

## Commands

```
cd shadow_frontier
# word -> integer (paste the letters as one string)
python3 -c "import sys; sys.path.insert(0,'search'); \
  from frontier import realize; \
  w=[int(c) for c in '111411211121...']; \
  r,Mb=realize(w); print(r, r.bit_length(), Mb)"
# integer -> word, plus the four-panel candidate image
python3 search/candidate_image.py --word 1,1,1,4,1,1,2,... --out viz/cand.png
python3 search/candidate_image.py --seed 687871 --steps 55 --out viz/c687871.png
```

## Worked example: catalog word #1

For the word #1 above ($M=240$, mean 1.583, $|D_3|=0.214$): $B_M=380$, modulus $2^{381}$, and the least positive representative is the 381-bit integer

45276335476484047115725251008950986904354815513999907611
28044450220436219657540681487193488995285987515743750727311

whose first 240 Syracuse letters reproduce the word exactly (machine-checked). Its fill ratio is $r$'s bit length over $B_M+1$: $381/381=1.000$ — and the half-word prefix already needs 184 bits of 185: the representative *grows with the prefix* ($184\to381$ bits from $M=120$ to 240) instead of stabilizing. That diagonal ride is the ghost signature of the Conventions section: the class contains no positive integer below $2^{380}$. Contrast a true integer realization: the survival champion 687871 *is* the least representative of its own 50-letter class, and stays put as the prefix grows — stabilization versus diagonal is the whole test.

# 3 The 120 pilot candidates: statistics

All 120 are ghosts: no representative stabilized; fill ratios $\approx 1$.

| # | mean $b$ | $\lvert D_3 \rvert$ | $P(16)$ | $B_{240}$ | rep bits | mod bits | fill |
|---|---|---|---|---|---|---|---|
| 1 | 1.5833 | 0.214 | 225 | 380 | 381 | 381 | 1.000 |
| 2 | 1.5792 | 0.214 | 225 | 379 | 379 | 380 | 0.997 |
| 3 | 1.5792 | 0.214 | 225 | 379 | 378 | 380 | 0.995 |
| 4 | 1.5833 | 0.239 | 205 | 380 | 381 | 381 | 1.000 |
| 5 | 1.5917 | 0.220 | 213 | 382 | 383 | 383 | 1.000 |
| 6 | 1.5792 | 0.215 | 225 | 379 | 379 | 380 | 0.997 |
| 7 | 1.5917 | 0.215 | 225 | 382 | 383 | 383 | 1.000 |
| 8 | 1.5792 | 0.216 | 225 | 379 | 380 | 380 | 1.000 |
| 9 | 1.5792 | 0.229 | 214 | 379 | 380 | 380 | 1.000 |
| 10 | 1.5792 | 0.233 | 225 | 379 | 380 | 380 | 1.000 |
| 11 | 1.5792 | 0.229 | 225 | 379 | 379 | 380 | 0.997 |
| 12 | 1.5792 | 0.216 | 225 | 379 | 380 | 380 | 1.000 |
| 13 | 1.5917 | 0.233 | 193 | 382 | 382 | 383 | 0.997 |
| 14 | 1.5792 | 0.215 | 224 | 379 | 380 | 380 | 1.000 |
| 15 | 1.5917 | 0.216 | 208 | 382 | 383 | 383 | 1.000 |
| 16 | 1.5833 | 0.214 | 225 | 380 | 378 | 381 | 0.992 |
| 17 | 1.5917 | 0.215 | 225 | 382 | 380 | 383 | 0.992 |
| 18 | 1.5792 | 0.218 | 225 | 379 | 380 | 380 | 1.000 |
| 19 | 1.5917 | 0.220 | 225 | 382 | 383 | 383 | 1.000 |
| 20 | 1.5792 | 0.260 | 214 | 379 | 380 | 380 | 1.000 |
| 21 | 1.5792 | 0.272 | 208 | 379 | 380 | 380 | 1.000 |
| 22 | 1.5917 | 0.220 | 214 | 382 | 383 | 383 | 1.000 |
| 23 | 1.5875 | 0.228 | 225 | 381 | 382 | 382 | 1.000 |
| 24 | 1.5875 | 0.246 | 214 | 381 | 380 | 382 | 0.995 |
| 25 | 1.5917 | 0.222 | 211 | 382 | 383 | 383 | 1.000 |
| 26 | 1.5917 | 0.242 | 189 | 382 | 383 | 383 | 1.000 |
| 27 | 1.5792 | 0.222 | 222 | 379 | 376 | 380 | 0.989 |
| 28 | 1.5917 | 0.214 | 211 | 382 | 383 | 383 | 1.000 |
| 29 | 1.5875 | 0.215 | 225 | 381 | 382 | 382 | 1.000 |
| 30 | 1.5792 | 0.260 | 217 | 379 | 375 | 380 | 0.987 |
| 31 | 1.5917 | 0.214 | 216 | 382 | 382 | 383 | 0.997 |
| 32 | 1.5833 | 0.239 | 203 | 380 | 380 | 381 | 0.997 |
| 33 | 1.5792 | 0.218 | 225 | 379 | 377 | 380 | 0.992 |
| 34 | 1.5875 | 0.217 | 225 | 381 | 380 | 382 | 0.995 |
| 35 | 1.5917 | 0.214 | 224 | 382 | 379 | 383 | 0.990 |
| 36 | 1.5917 | 0.216 | 225 | 382 | 377 | 383 | 0.984 |
| 37 | 1.5792 | 0.220 | 225 | 379 | 380 | 380 | 1.000 |
| 38 | 1.5917 | 0.225 | 225 | 382 | 383 | 383 | 1.000 |
| 39 | 1.5875 | 0.221 | 225 | 381 | 381 | 382 | 0.997 |
| 40 | 1.5792 | 0.218 | 225 | 379 | 379 | 380 | 0.997 |
| 41 | 1.5917 | 0.225 | 206 | 382 | 380 | 383 | 0.992 |
| 42 | 1.5792 | 0.238 | 212 | 379 | 377 | 380 | 0.992 |
| 43 | 1.5792 | 0.215 | 225 | 379 | 375 | 380 | 0.987 |
| 44 | 1.5917 | 0.215 | 225 | 382 | 383 | 383 | 1.000 |
| 45 | 1.5875 | 0.217 | 225 | 381 | 379 | 382 | 0.992 |
| 46 | 1.5792 | 0.215 | 219 | 379 | 378 | 380 | 0.995 |
| 47 | 1.5792 | 0.229 | 223 | 379 | 380 | 380 | 1.000 |
| 48 | 1.5792 | 0.215 | 225 | 379 | 377 | 380 | 0.992 |
| 49 | 1.5917 | 0.215 | 225 | 382 | 383 | 383 | 1.000 |
| 50 | 1.5917 | 0.216 | 225 | 382 | 382 | 383 | 0.997 |

| # | mean $b$ | $\|D_3\|$ | $P(16)$ | $B_{240}$ | rep bits | mod bits | fill |
|---:|---:|---:|---:|---:|---:|---:|---:|
| 51 | 1.5792 | 0.229 | 223 | 379 | 377 | 380 | 0.992 |
| 52 | 1.5792 | 0.248 | 206 | 379 | 380 | 380 | 1.000 |
| 53 | 1.5917 | 0.218 | 224 | 382 | 376 | 383 | 0.982 |
| 54 | 1.5792 | 0.215 | 225 | 379 | 380 | 380 | 1.000 |
| 55 | 1.5917 | 0.233 | 225 | 382 | 382 | 383 | 0.997 |
| 56 | 1.5917 | 0.279 | 194 | 382 | 383 | 383 | 1.000 |
| 57 | 1.5792 | 0.215 | 225 | 379 | 380 | 380 | 1.000 |
| 58 | 1.5792 | 0.218 | 225 | 379 | 377 | 380 | 0.992 |
| 59 | 1.5792 | 0.238 | 201 | 379 | 380 | 380 | 1.000 |
| 60 | 1.5792 | 0.214 | 225 | 379 | 380 | 380 | 1.000 |
| 61 | 1.5792 | 0.238 | 225 | 379 | 380 | 380 | 1.000 |
| 62 | 1.5833 | 0.230 | 213 | 380 | 381 | 381 | 1.000 |
| 63 | 1.5792 | 0.222 | 219 | 379 | 379 | 380 | 0.997 |
| 64 | 1.5792 | 0.215 | 219 | 379 | 379 | 380 | 0.997 |
| 65 | 1.5792 | 0.222 | 225 | 379 | 380 | 380 | 1.000 |
| 66 | 1.5917 | 0.216 | 196 | 382 | 383 | 383 | 1.000 |
| 67 | 1.5917 | 0.215 | 225 | 382 | 383 | 383 | 1.000 |
| 68 | 1.5792 | 0.216 | 225 | 379 | 372 | 380 | 0.979 |
| 69 | 1.5917 | 0.216 | 225 | 382 | 382 | 383 | 0.997 |
| 70 | 1.5917 | 0.216 | 225 | 382 | 383 | 383 | 1.000 |
| 71 | 1.5917 | 0.214 | 224 | 382 | 382 | 383 | 0.997 |
| 72 | 1.5792 | 0.214 | 225 | 379 | 380 | 380 | 1.000 |
| 73 | 1.5875 | 0.221 | 225 | 381 | 379 | 382 | 0.992 |
| 74 | 1.5917 | 0.215 | 222 | 382 | 383 | 383 | 1.000 |
| 75 | 1.5792 | 0.215 | 225 | 379 | 380 | 380 | 1.000 |
| 76 | 1.5792 | 0.225 | 225 | 379 | 380 | 380 | 1.000 |
| 77 | 1.5833 | 0.241 | 225 | 380 | 381 | 381 | 1.000 |
| 78 | 1.5917 | 0.215 | 225 | 382 | 382 | 383 | 0.997 |
| 79 | 1.5875 | 0.221 | 225 | 381 | 380 | 382 | 0.995 |
| 80 | 1.5875 | 0.217 | 225 | 381 | 381 | 382 | 0.997 |
| 81 | 1.5917 | 0.225 | 220 | 382 | 380 | 383 | 0.992 |
| 82 | 1.5833 | 0.214 | 222 | 380 | 381 | 381 | 1.000 |
| 83 | 1.5917 | 0.215 | 225 | 382 | 381 | 383 | 0.995 |
| 84 | 1.5917 | 0.218 | 225 | 382 | 383 | 383 | 1.000 |
| 85 | 1.5792 | 0.222 | 225 | 379 | 378 | 380 | 0.995 |
| 86 | 1.5792 | 0.238 | 225 | 379 | 380 | 380 | 1.000 |
| 87 | 1.5792 | 0.229 | 225 | 379 | 376 | 380 | 0.989 |
| 88 | 1.5875 | 0.227 | 225 | 381 | 382 | 382 | 1.000 |
| 89 | 1.5792 | 0.218 | 225 | 379 | 380 | 380 | 1.000 |
| 90 | 1.5917 | 0.216 | 225 | 382 | 383 | 383 | 1.000 |
| 91 | 1.5792 | 0.216 | 225 | 379 | 378 | 380 | 0.995 |
| 92 | 1.5833 | 0.225 | 225 | 380 | 381 | 381 | 1.000 |
| 93 | 1.5917 | 0.220 | 220 | 382 | 382 | 383 | 0.997 |
| 94 | 1.5792 | 0.214 | 225 | 379 | 380 | 380 | 1.000 |
| 95 | 1.5792 | 0.238 | 223 | 379 | 380 | 380 | 1.000 |
| 96 | 1.5792 | 0.238 | 225 | 379 | 379 | 380 | 0.997 |
| 97 | 1.5917 | 0.233 | 221 | 382 | 383 | 383 | 1.000 |
| 98 | 1.5792 | 0.229 | 219 | 379 | 380 | 380 | 1.000 |
| 99 | 1.5917 | 0.218 | 223 | 382 | 382 | 383 | 0.997 |
| 100 | 1.5792 | 0.215 | 225 | 379 | 379 | 380 | 0.997 |
| 101 | 1.5792 | 0.218 | 225 | 379 | 380 | 380 | 1.000 |
| 102 | 1.5792 | 0.218 | 216 | 379 | 376 | 380 | 0.989 |
| 103 | 1.5792 | 0.229 | 225 | 379 | 379 | 380 | 0.997 |
| 104 | 1.5875 | 0.217 | 225 | 381 | 382 | 382 | 1.000 |

| # | mean $b$ | $\lvert D_3\rvert$ | $P(16)$ | $B_{240}$ | rep bits | mod bits | fill |
|---|---|---|---|---|---|---|---|
| 105 | 1.5917 | 0.229 | 211 | 382 | 383 | 383 | 1.000 |
| 106 | 1.5833 | 0.232 | 225 | 380 | 380 | 381 | 0.997 |
| 107 | 1.5792 | 0.218 | 225 | 379 | 379 | 380 | 0.997 |
| 108 | 1.5792 | 0.214 | 225 | 379 | 380 | 380 | 1.000 |
| 109 | 1.5917 | 0.220 | 225 | 382 | 381 | 383 | 0.995 |
| 110 | 1.5917 | 0.220 | 225 | 382 | 383 | 383 | 1.000 |
| 111 | 1.5792 | 0.229 | 225 | 379 | 380 | 380 | 1.000 |
| 112 | 1.5833 | 0.218 | 225 | 380 | 379 | 381 | 0.995 |
| 113 | 1.5792 | 0.248 | 225 | 379 | 379 | 380 | 0.997 |
| 114 | 1.5792 | 0.229 | 223 | 379 | 380 | 380 | 1.000 |
| 115 | 1.5917 | 0.220 | 224 | 382 | 383 | 383 | 1.000 |
| 116 | 1.5917 | 0.233 | 200 | 382 | 382 | 383 | 0.997 |
| 117 | 1.5792 | 0.222 | 225 | 379 | 378 | 380 | 0.995 |
| 118 | 1.5875 | 0.214 | 225 | 381 | 382 | 382 | 1.000 |
| 119 | 1.5792 | 0.218 | 224 | 379 | 380 | 380 | 1.000 |
| 120 | 1.5792 | 0.214 | 225 | 379 | 380 | 380 | 1.000 |

# 4 Sturmian ghosts (critical slope, now a theorem)

Words $b_n = 1 + \mathbf{1}\{\{n\alpha + \theta\} \geq \alpha\}$, $\alpha = 2 - \log_2 3$. By the repetition-rigidity theorem these are non-realizable for *every* slope and intercept; the data below shows the same diagonal-riding signature as the candidates.

| $\theta$ | $\lvert D_3\rvert$ | $B_{240}$ | rep bits | mod bits | fill | rep bits at $N$=100/200 |
|---|---|---|---|---|---|---|
| 0.000 | 0.447 | 380 | 381 | 381 | 1.000 | 159/316 |
| 0.100 | 0.447 | 380 | 380 | 381 | 0.997 | 157/318 |
| 0.200 | 0.447 | 380 | 377 | 381 | 0.990 | 159/317 |
| 0.330 | 0.447 | 380 | 381 | 381 | 1.000 | 155/318 |
| 0.500 | 0.453 | 381 | 379 | 382 | 0.992 | 159/317 |
| 0.610 | 0.453 | 381 | 376 | 382 | 0.984 | 160/318 |
| 0.750 | 0.453 | 381 | 382 | 382 | 1.000 | 160/317 |
| 0.900 | 0.447 | 380 | 381 | 381 | 1.000 | 160/318 |
| 0.057 | 0.447 | 380 | 381 | 381 | 1.000 | 159/318 |
| 0.085 | 0.447 | 380 | 381 | 381 | 1.000 | 159/318 |

Example Sturmian words (first 120 letters):

$\theta = 0.00$:

```
122121221212212121221212212121221212212121221212212122121212
212122121212212122121212212122121212212122121221212122121221
```

$\theta = 0.10$:

```
122121212212122121221212122121221212122121221212122121221212
122121221212212121221212212121221212212121221212212122121212
```

$\theta = 0.20$:

```
121221212212121221212212122121212212122121212212122121212212
122121212212122121221212122121221212122121221212122121221212
```

$\theta = 0.33$:

```
121221212122121221212122121221212122121221212122121221212212
121221212212121221212212121221212212121221212212122121212212
```

## 5 The negative cycles, computed explicitly

The three known negative cycles anchor the ghost families: each periodic valuation word below has its unique 2-adic realization at a *negative* integer, so the least positive representatives of its realizability classes ride the ceiling $2^{B_N+1} - |x|$ — the periodic ghost lines of the trajectory zoo. Throughout, $S(x) = (3x+1)/2^{v_2(3x+1)}$ and a period-$p$ word $w$ with sum $B$ satisfies the affine cycle equation

$$x = \frac{A_w}{2^B - 3^p}, \qquad A_w = \sum_{j=0}^{p-1} 3^{\,p-1-j}\, 2^{B_j}, \quad B_j = b_1 + \cdots + b_j.$$

Negative cycles are exactly those with $2^B < 3^p$ (denominator negative).

### $x = -1$: word $(1)$, period $1$

$$3(-1) + 1 = -2 = -2^1 \cdot 1, \qquad v_2 = 1, \qquad S(-1) = \tfrac{-2}{2} = -1.$$

Cycle formula: $p = 1$, $B = 1$, $A_w = 3^0 2^0 = 1$, and $x = 1/(2^1 - 3^1) = 1/(-1) = -1$. This is the 2-adic limit of the all-ones word: $b \equiv 1$ forces $x \equiv -1 \pmod{2^k}$ for every $k$.

### $x = -5$: word $(1, 2)$, period $2$

$$\begin{aligned} 3(-5) + 1 &= -14 = -2^1 \cdot 7, &\qquad v_2 &= 1, &\qquad S(-5) &= -7,\\ 3(-7) + 1 &= -20 = -2^2 \cdot 5, & v_2 &= 2, & S(-7) &= -5. \end{aligned}$$

Cycle formula: $p = 2$, $B = 1 + 2 = 3$, partial sums $B_0 = 0$, $B_1 = 1$:

$$A_w = 3^1 2^0 + 3^0 2^1 = 3 + 2 = 5, \qquad x = \frac{5}{2^3 - 3^2} = \frac{5}{8 - 9} = -5.$$

Letter mean $\frac{3}{2} < \log_2 3$: in the positive world this word ascends, the gentle green dotted line of the trajectory zoo.

### $x = -17$: word $(1, 1, 1, 2, 1, 1, 4)$, period $7$

$$\begin{aligned} 3(-17) + 1 &= -50 = -2^1 \cdot 25, &\qquad S\colon -17 &\mapsto -25,\\ 3(-25) + 1 &= -74 = -2^1 \cdot 37, & -25 &\mapsto -37,\\ 3(-37) + 1 &= -110 = -2^1 \cdot 55, & -37 &\mapsto -55,\\ 3(-55) + 1 &= -164 = -2^2 \cdot 41, & -55 &\mapsto -41,\\ 3(-41) + 1 &= -122 = -2^1 \cdot 61, & -41 &\mapsto -61,\\ 3(-61) + 1 &= -182 = -2^1 \cdot 91, & -61 &\mapsto -91,\\ 3(-91) + 1 &= -272 = -2^4 \cdot 17, & -91 &\mapsto -17. \end{aligned}$$

Cycle formula: $p = 7$, $B = 1{+}1{+}1{+}2{+}1{+}1{+}4 = 11$, so $2^B = 2048 < 2187 = 3^p$; partial sums $(B_0, \ldots, B_6) = (0, 1, 2, 3, 5, 6, 7)$:

$$A_w = 3^6{\cdot}1 + 3^5{\cdot}2 + 3^4{\cdot}4 + 3^3{\cdot}8 + 3^2{\cdot}32 + 3{\cdot}64 + 128 = 729 + 486 + 324 + 216 + 288 + 192 + 128 = 2363,$$

$$x = \frac{2363}{2^{11} - 3^7} = \frac{2363}{2048 - 2187} = \frac{2363}{-139} = -17 \qquad (139 \cdot 17 = 2363).$$

Letter mean $\frac{11}{7} \approx 1.571 < \log_2 3 \approx 1.585$: again sub-critical, again a positive-world ascender. Note how close $11/7$ is to $\log_2 3$ — the $-17$ cycle is a near-balanced word, the best rational approximant realizable at small height, and exactly the shape that the Baker cycle bound ($x_{\min} \le Cp^{\tau+1}$) governs.

# 6 The June 2026 constraints, and the surviving candidate list

Since this catalog was first compiled, the program proved a stack of further theorems (companion notes: narrow-band exclusion; two clocks; universal ghost clock; CIC round chain; faithfulness). This section records what they jointly do to the counterexample profile, and then lists what now survives.

## The new constraints, in catalog form

- *Exact narrow-band exclusion.* No acyclic orbit dwells more than $CY^{\theta(D)}$ consecutive steps in a band $[Y, 2^{2D}Y]$; the count of confined words is an exact integer (no Perron estimates), and the counting capacity is exhausted at $D^* = 1.246$ (band ratio 5.63). Bounded-discrepancy words ($\sup |S_j| < 1.246$) are unconditionally non-realizable by divergent orbits — the Sturmian section above is now a corollary. In a ratio-$\leq 2$ band the dwell bound is *logarithmic* (the dyadic strip carries one word per phase).

- *The phase diagram has two arithmetic lines.* Vertical: $D^* = 1.246$. Horizontal: the binary bias ceiling $\varepsilon_{\text{bin}} = 0.449$ — calibrated $\{1, 2\}$-words cannot carry more order-3 shadow bias than this; exceeding it requires letters $b \geq 3$, which are descent-heavy. Saturation $\varepsilon^* \approx 0.54$.

- *Every letter is a metered ghost approach.* $b$=1 runs meter $-1$ (length exactly $v_2(x+1)-1$); $b$=2 runs meter the trivial cycle $+1$ (depth decrement exactly 2, parity handoff); $b \geq 3$ letters read $v_2(x + \frac{1}{3})$. Universally: every signature $\sigma$ of depth $K$ has phantom root $\rho_\sigma = C_\sigma/(2^K - 3^\ell)$ (the cycles above are instances), alignment drops exactly $K$ per ridden block, rides last exactly $\lfloor (a-1)/K \rfloor$ blocks, and alignment $a$ at height $x$ costs $2^a \leq |2^K - 3^\ell| x + C_\sigma$. Amortized over *all* ghost families of depth $\geq 3$, the available drift gain is $R = 0.0882$ bits per letter against a 0.415-bit budget.

- *Supply is consumed without replacement.* An acyclic orbit visits each integer once: each dyadic band holds only $2^{k-g}$ points of depth $g$ per ghost, ever. Consequences: at most $2^{k-3}$ upward crossings of level $2^k$; a divergent orbit spends $O(X \log X)$ lifetime steps below $X$.

- *The survival coin.* Parsed at 1-run starts, life against the letters-$\leq 2$ regime is the reading $v_2(3^\ell m - 1) = v_2(m - 3^{-\ell})$, death iff odd. On Haar fibers the coin is exactly i.i.d. fair-thirds; on ghost-aligned seeds it is deterministic (LTE): Mersenne $2^g - 1$ survives round one iff $g$ is odd with $v_2(g-1)$ even; the 3-smooth families $2^g 3^a \mp 1$ cascade in closed form, with round-two readings such as $v_2(a' + 2 - \mathrm{Log}_3(-5))$ — the 2-adic *logarithm of the* $-5$ *ghost above* — and every tested seed dies by round 7.

- *The* $\{1, 2\}$ *alphabet has an entropy deficit.* $\theta_{12}(D) \leq 0.97907 < 1$ at every band width: for an orbit with no letter $\geq 3$ the phase diagram's allowed region is empty — no band of any ratio offers refuge.

- *The blind spot bites.* A positive integer has $\log_2 x_0$ bits, period: after $\approx \frac{2}{3} \log_2 x_0$ steps the word has consumed them all and every later letter is forced by the $\times 3$ carry cascade. The infinite word has Kolmogorov complexity $\leq \log_2 x_0 + O(1)$ while repetition rigidity demands near-maximal factor complexity in banded windows. Realizable words are a countable, computable family (of the $2^{30}$ depth-30 words over $\{1, 2\}$, exactly 66 are realized below $10^6$).

## The composite profile

A counterexample must therefore be: a divergent orbit, fleeing upward (escape $O(X \log X)$, single excursions below $X$ sublinear), never pausing in any narrow band, calibrated to mean $\log_2 3$ yet carrying non-coboundary bias below the ceilings (0.449 binary, $\approx 0.54$ saturated), financing every ascent through without-replacement ghost-depth pools and an amortized 0.088-bit census margin, beating an i.i.d. fair-thirds coin forever on its +1-handoff parities, with a word of near-maximal local diversity generated — after a free opening of $\frac{2}{3} \log_2 x_0$ moves — by a fixed finite carry transducer with no further choices.

## The candidate list, June 2026

No specific integer candidate exists: every concrete object ever computed in this program is either a 2-adic ghost or dies with a finite certificate. What survives is six structural profiles. For each we list its best known approximant and the meter that taxes it.

| profile | best known approximants | taxing mechanism / status |
|---|---|---|
| C1. Carry-cascade pseudo-random divergent (generic profile: calibrated, biased, high-complexity) | the 120 pilot words of this catalog (all ghosts); integer side: survival champions 27, 4591, 31911, 517191, 687871 (depths $17, 27, 33, 38, 50$; 59–76% of each life is in the forced carry regime) | fair-thirds survival coin; census margin 4.7×; all known members die; champion height grows $\approx 2^{D/2.2}$ with depth |
| C2. Binary violator: letters $\{1, 2\}$ forever ($\mathcal{C}_\infty \cap \mathbb{N}$, the CIC class) | none positive; the 2-adic members are the Sturmian-type ghosts of §3 and their high-complexity cousins | entropy deficit $\theta_{12} < 0.9791$: no band of any width is a refuge; parity coin must read even forever |
| C3. Parity-channel rider (word biased toward $(1, 1, 2)^\infty$) | integers 2-adically near $-19/11$; least representatives ride the full diagonal (ghosts) | the $K$=4 clock of $-19/11$: alignment $a$ costs height $11x + 19 \geq 2^a$, rides last $\lfloor (a-1)/4 \rfloor$ blocks |
| C4. Ghost-aligned seeds (3-smooth neighborhoods $2^g 3^a \mp 1$, incl. Mersenne) | round-one survivors $2^5-1, 2^{13}-1, 2^{17}-1, \ldots$ ($g - 1 \equiv 2^{2t} \bmod 2^{2t+1}$); family champion $2^{705} - 1$ ($n = 704$: six rounds, readings $8, 2, 2, 2, 2, 2, 3$) | deterministic LTE cascade; ghost-log readings; every tested seed ($n \leq 3000$) certified dead by round 7 |
| C5. High cycle (near-balanced periodic word, $B/p$ a convergent of $\log_2 3$: $19/12$, $65/41$, $84/53$, $485/306, \ldots$) | the $-1$, $-5$, $-17$ cycles above (negative side); no positive candidate below current $k$-cycle exclusions | cycle formula $x = A_w/(2^B - 3^p)$ + Baker pinning $x_{\min} \leq Cp^{\tau+1}$ + narrow-band period localization |
| C6. Exponent-level conspirator (round-chain exponents tracking 2-adic digits of ghost logarithms, e.g. $N_0 = \mathrm{Log}_3(-5) = 2556611305511861515 \bmod 2^{64}$) | none known; alignment at exponent level costs height at exponent level | the lifted clock: ghosts-of-ghosts; $p$-adic Baker bounds the readings effectively |

The list should be read as a specification of where a counterexample would have to hide, not as evidence that one exists: profiles C2–C6 are each pinned by an exact meter with a positivity cap, and C1 — the generic impostor — is the quenched calibration wall itself, the single open core to which every known framework (ours, Tao's, the LLM-collaboration manuscript audited in the companion notes) independently reduces.

# 7 Interlude: could a counterexample carry a message?

A speculation, wild but within reason. The counterexample signature is near-critical drift + persistent non-coboundary bias + high local word complexity + mixed $(2, 3)$-adic coherence. Suppose someone wanted to *hide a message* in such an object. Could word complexity communicate information?

## The channel, and its exact capacity

Any finite message can be written into an orbit's opening. Each scripted letter costs $\approx \log_2 3 = 1.585$ bits of seed (the cylinder modulus of §2), so a seed of $n$ bits buys a free composition of $\approx 0.63\,n$ letters — and then the blind spot takes over: the class pins the seed, and *every subsequent letter is forced by the* $\times 3$ *carry cascade*. The total information content of an entire infinite Collatz word is $K(\text{word}) \leq \log_2 x_0 + O(1)$: a counterexample can only ever say one thing — its own name. The survival champions live this: 59–76% of their lives are in the forced regime, improvisation on a theme they no longer control.

Worked example: the 56-bit integer

```
60822829639107739
```

has a Syracuse orbit whose first 40 valuations spell GHOST in ASCII ($b{=}1 \mapsto 0$, $b{=}2 \mapsto 1$), at 1.475 seed-bits per scripted letter.

## The Kraft refinement: why 56 bits for a 40-bit message

The cost of a letter is not one bit but $b$ bits: the constraint $v_2(3x+1) = b$ pins $b$ binary digits of the seed (plus one overall). In the naive scheme a message-bit 0 costs 1 seed-bit and a message-bit 1 costs 2; GHOST's 40 bits hold 18 ones and 22 zeros, so $B + 1 = 22 + 2 \cdot 18 + 1 = 59$, and the printed integer is the least member of a $2^{59}$-modulus class that happened to start with three zero bits (a $\sim 1/8$ accident): 40 message bits, 59 bits of congruence conditions, 56-bit smallest solution.

The cost structure — one letter per cost class $b$, cost $b$ — satisfies Kraft with *equality*, $\sum_b 2^{-b} = 1$: *the Syracuse alphabet is a complete binary prefix code*, which is the lossless-coding conjugacy of the faithfulness note in information-theoretic clothes. The optimal encoder therefore maps the letter $b$ to the bit-block $1^{b-1}0$ (pad the message with a final 0): each letter then carries exactly as many message bits as it costs, and the rate is 1 seed-bit per message-bit. For GHOST this gives 23 letters (largest letter 5), $B + 1 = 42$, and the performing integer drops to the 42-bit

```
3925048663003,
```

1.050 seed-bits per message-bit — essentially capacity (the excess is the padding zero and the +1 in the modulus). Read backwards, the same identity is why nothing beyond the seed can ever be communicated: the letter stream is a rate-1 recoding of the seed's binary digits through the carry transducer; the channel runs at capacity before you have said anything.

## The information paradox

To survive, the stream must *look like* a random source of about 1.4 bits per letter (the calibrated entropy ceiling, less the Legendre tax $t^*$ once the bias exceeds the free basin $\varepsilon_{\text{nat}} \to 0.1397$), with near-maximal factor diversity in every banded window at scale $p \approx \log_2 Y$, forever — repetition rigidity tolerates nothing less. Yet its true information rate is asymptotically *zero*: finite seed, infinite output. By the completeness theorems of the faithfulness note, the statistical tests available to the dynamics are exactly the finite-shadow algebra; so a counterexample is a pseudorandom generator engineered to *fail precisely one named test* (the chosen non-coboundary character, where it must show constant bias) *while passing all the others* — built not by a cryptographer but by carry propagation. The budgets are quantitative: bias up to 0.1397 is free; every structured ghost channel combined supplies at most $R = 0.0882$ bits per letter of the 0.415 needed ($\leq 21\%$ efficiency); the rest must come from the unstructured remainder where no finite certificate sees it.

## How such a word would function

A self-feeding stream cipher. State: the binary tape of $x_n$. Output: the $v_2$-readings at the bottom. Update: multiply by 3, propagate carries upward, truncate what was read. The opening $0.63 \log_2 x_0$ moves are the composer's; afterwards the machine's "randomness" is the binary digit stream of 3-power multiples of its own past — it reads its exhaust as fuel. By the round chain, each life-or-death reading is $v_2(3^\ell m - 1)$, so the functional requirement is that *the powers of* 3*, in base* 2*, along this one orbit, are non-normal in a precisely coordinated pattern.* That is the entire miracle, localized — the regime where mathematics has only logarithmic digit bounds (Stewart), and exactly where the program's wall stands. A counterexample would be a finite phrase whose forced elaboration is an unending, never-self-quoting, undetectably biased improvisation: a message that continues writing itself after its author has run out of ink.

## Play with it

Spell any ASCII message into an orbit and read it back (from the project root `shadow_frontier/`):

```
python3 - <<'EOF'
import sys; sys.path.insert(0,'search')
from frontier import realize
```

```
def spell(msg):                         # naive: 0->b=1, 1->b=2
    bits = ''.join(f"{ord(c):08b}" for c in msg)
    word = [1 if c == '0' else 2 for c in bits]
    return realize(word)[0]

def spell_kraft(msg):                   # optimal: letter b <-> 1^(b-1) 0
    bits = ''.join(f"{ord(c):08b}" for c in msg) + '0'
    word, run = [], 0
    for c in bits:
        if c == '1': run += 1
        else: word.append(run + 1); run = 0
    return realize(word)[0]

def read_kraft(x, nchars):              # integer -> message
    out = []
    while len(''.join(out)) < 8 * nchars:
        t = 3*x + 1; b = (t & -t).bit_length() - 1
        out.append('1' * (b - 1) + '0'); x = t >> b
    s = ''.join(out)[:8 * nchars]
    return ''.join(chr(int(s[i:i+8], 2)) for i in range(0, len(s), 8))

print(spell("GHOST"))            # 60822829639107739  (56 bits)
r = spell_kraft("GHOST")
print(r)                         # 3925048663003      (42 bits)
print(read_kraft(r, 5))          # GHOST
EOF

# render the four-panel candidate image of your message-orbit:
python3 search/candidate_image.py --seed 60822829639107739 \
    --steps 40 --out viz/cand_ghost_msg.png
```

Things to try: longer messages (cost $\approx 11.8$ seed bits per character — `spell` of a 100-character message returns a $\approx 1180$-bit integer, instantly); watch the height panel during the scripted prefix (your bit pattern *is* the drift: letters 1 ascend, 2 descend, so the orbit's silhouette draws your message's bit-density); read past the end of your message to watch the carry cascade take over the composition — the letters after $8 \cdot |\text{msg}|$ are the automaton's, not yours, and (the $(3/4)^D$ law) a random message's orbit takes a letter $b \geq 3$ within a handful of steps of being left to itself; try to compose a message whose forced continuation survives long — that is exactly the champion-hunting problem, and the house always wins.

We played the game (`search/message_champions.py`). Over the 44,000-word English dictionary the champion is `bopped`: 48 scripted letters, then the cascade composes 36 *more* on its own (integer 1769681963780500439019); runner-up `Bancroft` (+36 after 64). The best two-character message is `a‘`: a 7-digit seed (4063723) whose 16-letter script the automaton extends by +34 — total 50 letters from 22 bits, ratio 2.3, i.e. a survival champion wearing a two-character name. And the law is exact: best-of-$N$ search should win $\log_{4/3} N$ forced letters — prediction 37 for the dictionary ($N$=44k), 32 for two characters ($N$=9025); measured 36 and 34. The dictionary behaves like 44,000 random seeds: there are no magic words, only Borel–Cantelli paying out at its stated rate.

## 8 Collatz Elevator Pitch

One-liner. "We can prove the average number loses this game. We can't prove that this particular number does — and every number is a particular number."

Thirty seconds. "The rule is simple: take a number — if it's odd, triple it and add one; if it's even, halve it. The conjecture says everyone eventually crashes down to 1. Now, the ups and downs of any number look exactly like coin flips, and the coin is rigged downward — so on average, everything falls. That part is basically understood. The problem is that nothing here is actually a coin flip. Each number's entire future — every up, every down, forever — was fixed the moment you chose it, written into its digits. So to prove the conjecture, you'd have to rule out that somewhere out there is one number whose predetermined script just happens to dodge the crash forever. Probability can't do that: probability talks about most numbers, never about that one. It's the same wall as: we can prove almost every number has perfectly shuffled digits, but nobody can prove $\sqrt{2}$ does — even though $\sqrt{2}$ is sitting right there, fully determined."

The twist, if she wants one more floor. "Here's what makes it maddening — and what the last year of work actually showed. We've cornered the hypothetical escape artist. It can't follow any pattern: we proved that if its moves ever repeat, even once, in the wrong circumstances, that repetition snaps shut into a loop, and all loops are dead. So a survivor would have to move in a way that's indistinguishable from pure chance — never repeating, perfectly balanced on the knife edge between rising and falling. But it also can't actually be random, because random walkers provably crash. It would have to be a perfect impostor: a fully scripted number flawlessly impersonating chance, forever, while hiding a secret bias it needs to stay alive. We're fairly sure that impostor can't exist — being biased and looking unbiased pull against each other — but proving a specific, predetermined thing can't perfectly imitate randomness is something mathematics doesn't yet know how to do. That's the whole obstruction, in one sentence: we can beat every strategy, but we can't beat 'no strategy.'"

A note on your original phrasing: "deterministic numbers that behave totally randomly" invites the response "then they crash, like random things do —done!" The fix is to put the burden where it really sits: deterministic numbers that we can't prove aren't perfectly disguised as random. The casino version also lands well: "We've proven the house always wins on average. The conjecture asks us to prove no individual gambler beats the house forever — when every gambler's every move was decided before they walked in."

# A The 120 candidate words

Alphabet $\{1, 2, 4\}$, length 240, row-wrapped at 60 letters.

#1 (mean 1.583, $|D_3|$=0.214):
111411211121121111112114421114112222111122212112122212112221
112221122411212122222121212111211111121221222111411112211121
222111111112422222211111212114112121112142211141421211111141
211211412222121111111122222121141212112214111211111114422241

#2 (mean 1.579, $|D_3|$=0.214):
2222222222222222111112112114211112114211112212222221114241211
111112412111211422111141112111122112212212111111212111111211
212222111212111114221424111211121411211144212111114121114221
211211111111122121111121211221121211141121141242222122222111

#3 (mean 1.579, $|D_3|$=0.214):
222112111112112421221112111211111212122111211111211212221144412
4222222111221212111111121111111211412121112224222214111212111
112411112112222421111211121111221211221111241121111224111121221
2241221141121111221224121124211222121114411211112112121111122

#4 (mean 1.583, $|D_3|$=0.239):
11111111111111111111111111111111111111212221141142141221114212
111421141112214212421122114411224111121112212122112222221111
111112221221121111111111121421121124112212121142244122121211
1214114122111211112112211114224244212121121211421122221121111

#5 (mean 1.592, $|D_3|$=0.220):
1111111111111111111111111111121121111412414211121212411112242
121121121121211114211141222111242122111211222111111111122111
221222112412112111422412112222244212121411111124111221221212
212211211411111122112142111122211122224114212211221214211224

#6 (mean 1.579, $|D_3|$=0.215):
2222222242222222421121111141242121141122122222111112112241221
1111211211111111111111221111122222421114212214211111112422221
111112221111112111222214111122221121211242111111411121211222
111112111112111112211221214211141122121141214111121211212112

#7 (mean 1.592, $|D_3|$=0.215):
121112111211212121111224121112121411212114112411221242222211
221221111112211111221212224214111211122112111222111211112112
212422141111221412421111222214112142112111121212111112211111
111141212222242111111221211221111212421212114112211222141211

#8 (mean 1.579, $|D_3|$=0.216):
1221111111111212112111114112122221111121111211111211142211414
222221141211111421222112121111112114122121211122121211421121

4121211241122111124221122211111114114212111222224111121111221
2141221211141121112112211412241111121121124121212212112222221

#9 (mean 1.579, $|D_3|$=0.229):
2222222222222242222222222222222222222222242221112222112211144111
1211211221112211122111111211142124111122211121121111222241111
2412121211411111211121111212211111221112111141212121142111111
1111221122111111211111142242121121121111221111111111112112421

#10 (mean 1.579, $|D_3|$=0.233):
1111121111222121222121211114111414222441214211212124111122211
1111111111211214211114111222212112242114221411221441421111121
1421411112122111112122224111111111111111224221212122111111111
1112214221112121111111111422111211111212111211211121114122221

#11 (mean 1.579, $|D_3|$=0.229):
2222224222222422222221122112212412212211122111422111122122112
1111121111211212111121111221121121122112211112224112121124114 1
1212111142111211211221114121212112142221111121124112112 22121
2242212122122121121114212111214122111111122111111112122111111

#12 (mean 1.579, $|D_3|$=0.216):
4222112121141211222441111222211124112212211121211124411244 21
1122221111121111211122121112212212212111111111121242212112122
1141222141211111112412411112114112121111111411122221141211212
2111121112111111112224211211142111112122121211211111212211221

#13 (mean 1.592, $|D_3|$=0.233):
1111111111111111111111111111111111111111111111111222212111121
1221242141122111111122111112221221411144211222112124421141111
1212414211222241221114221212112224211221144211424221212211111
2111241442121221121212122211121112411121111211211142121 22141

#14 (mean 1.579, $|D_3|$=0.215):
2422222222422222221111111121122111111111111111111221121224211
2111212122221121221214121121221121221211212121111211421111121
2121112211141111211212441222112111221111221241241121122222121 2
2112111111112112141141111121241112241411241221212111112114 11

#15 (mean 1.592, $|D_3|$=0.216):
1111111111111111111111111111111111124411122111211111211221122 1
1212211211412412142112412211111112241211222222111112222111122
1222122221421212121422222411214111211211122221411414111122111
2112112211212421121141242112112241142122222111411222221111 11

#16 (mean 1.583, $|D_3|$=0.214):
2122141111121111212111112122111212111121122414224112111221122
2411221421212111111241211244411221221111111222121242222112 14
2111214211111121221111242112212114121111112211221111111121222
1141212114112111111241222121221211112121212212121121112111 22

#17 (mean 1.592, $|D_3|$=0.215):
1112112122421212222112212221111121224211121121122222114111212
2111112421224121121121211122111411221411211242211122221211411
4111114212112211121122111122111211111111221221121221121114112
2122211111111141114112112141111112221121242111211222424212 22

#18 (mean 1.579, $|D_3|$=0.218):
2222222242422111241111221221211411221221121212211221142411 12
1112242112211121211141111212114111112221211212124121111121222
1211412111122222111211111221141241411221112211112122122221111
2122142211211211111212121111112221114211211122111121111112211 1

#19 (mean 1.592, $|D_3|$=0.220):
1124111112111112121112411212111212211211111412211111211111222 1
1212121112122121121211141111111121411221224421211221122222112
2114142111411121122122441111112112111121122214124111412122 22
1111211221244112121111221212411212411212142422121122111111 11

#20 (mean 1.579, $|D_3|$=0.260):
2222224222222222222222222222222222221212211111221111112211241 12
2111121122112241211141111222112122121122211111114121222122 12
4211142121242212121121122111111211221111122111221242211112 11
1212111111212221211112111111111111114112111241121222122212 1111

#21 (mean 1.579, $|D_3|$=0.272):
2222222222222222222222222222222222221212111122122111211222221 21
1111112212112111122211112111121111122211121221111411214141 21
1112112111214222112112121211224212122111124142121111211211 21
1211211121111111221212112211211222122221244122121111114121 22

#22 (mean 1.592, $|D_3|$=0.220):
1111111111111111111111111111112111122421242111421241221111122 41
2211142142122111222212214144211211122211211412221112111112 212
1141111121114112121111211112244112121112211212224211121111 14
1211111212112121242422121212412122111224111112121221222112 12

#23 (mean 1.587, $|D_3|$=0.228):
142222112411411141112111121121122414422122111111211211222412
222121111211221211111211122114112121214111111211111221122122
124221212111111412142112112221111212111114121112421211112112
122411111111111111121111222221111224244111241142141111122111

#24 (mean 1.587, $|D_3|$=0.246):
111111111111111111111112114212211111111111111111111122122211412
121212122414211411111111214212211411121124212122222414114212
211414122212111221211121112111241111111221142111211112122141
212111142121111214124412121221222112221411121421141114211112

#25 (mean 1.592, $|D_3|$=0.222):
111111111111111111111111111111242212211112222112242111111411
121112221221112421142211241212122114111121122124112221241421
221112412222221212211122224224211212124211214222221211111122
211111121211221122122121112242221211122221111111122122112211

#26 (mean 1.592, $|D_3|$=0.242):
111111111111111111111111111111111111111111111111111121111111
2212222211111221422411122421111141111111221112144222421121124
214244121121122112141121212241411421221111411122211122141211
112112222442112212212412211111114124222122121141242212211211

#27 (mean 1.579, $|D_3|$=0.222):
224222222222222222222221111222111121211121111211111111211111111
122211221242222122121111111112214111212221211111421112141422
411121121111111121211222411411221122211111111122111212112112
221424122222421221111111111411112142222111111211422212121121

#28 (mean 1.592, $|D_3|$=0.214):
111111111111111111111111111111121114411214111212111211111121122
411214112111222122112212212222222422122412111212212112412111
112112121421211211141112211124211241122421222221122112122121
121211122211112112144241211122442221121211112212111412211212

#29 (mean 1.587, $|D_3|$=0.215):
112112414211111111212422111111111112111222111411211244211212
412112121222112122221211221211112211114211221222141212121124
111211112422112111242412111211122211111222121112212111121112
112212121211111442114414211211212112114211221212121222111111

#30 (mean 1.579, $|D_3|$=0.260):
222222222222222242222222222222222222222221411112112111111112211
221111111211111241112222121122414212212122114112121121212121122
111111112222111211121111111111121111111111442221212121414141221
212211211211122111111112121211121122224122121111122211222211

#31 (mean 1.592, $|D_3|$=0.214):
111111111111111111111111121122112211122114211122122222221214
212222111112121211122212112212222211111122122141241211211111
141121112111141114211442224211222241211111412111211441122221
112221421211111121112211242111224111122122412211222121421111

#32 (mean 1.583, $|D_3|$=0.239):
11111111111111111111111111111111111111111112111124112211421214211
241122242112114144211221211112111221121211121222112212122112
112211112111114121142142122114224221211214111221212111412212
221111112141221112111411444111421221241122111121112221224111

#33 (mean 1.579, $|D_3|$=0.218):
4212412221211211211121211212221111212411111111222212112112241
222111121111112141221124222122112111211411111222112211121111
111111212211111122111221122221244111141242121122112214221112
121221211222212111221112211114212222211112422211111111124412

#34 (mean 1.587, $|D_3|$=0.217):
114111211111111411112144142414111211211122111122211122111122
112111222222211221122121212211221122111111112111124211221211
112412122121222224122421211111111214111221242211211212211412
211221211212111211141111111111111411421411214222211121421121

#35 (mean 1.592, $|D_3|$=0.214):
111111111111111114122211121112121421442122121242212121241214
112112122421122111121411241222212242111221121222241111212122
111111121211212141212212112412412222111111411221112122241221
222111122111111211121112112241211222111112124111121112111111

#36 (mean 1.592, $|D_3|$=0.216):
1111111111111212122221212111112121221142214112122222121121211
112112112212111111211412114212111221121141111114111121124211
111142112211422122114111411224211112124141221422221124214121
212411111122112121111211411112122122122211122411212211212211

#37 (mean 1.579, $|D_3|$=0.220):
222222241111111112121124422121211111121111111114121212411224
12421111121112212121211142211111121211111111111122112414112212

1122112142111212242111221211111111112124124111111424112111111
2221211122421122412211112222124111121211122412112211112111121

#38 (mean 1.592, $|D_3|$=0.225):
1111111111111242112112121122212212111124112121111211124111112
1212222122442211222114242211211111222121141111211221211122142
1412144421111241242241211212111214212141121111221111111241121
2211111411121111211121111211111112211221142221111111121214112

#39 (mean 1.587, $|D_3|$=0.221):
2241112122211411111122412221114112111111114112121112412112111
1112111222212141111422114112211221122121112122121111121111111
2414112111212421212111422221111141211211211111212121121111111
1442212111121241221221112222111211211142212121111224442112211

#40 (mean 1.579, $|D_3|$=0.218):
2222222222211411221111112141112111121111121112111222411221111
1121214111112241112111122221221122111221121122121114121422122
2221221222212221121111211141122211212111121411112211222114111
1211112112124114122122122121212114112214412111111111242111212

#41 (mean 1.592, $|D_3|$=0.225):
1111111111111111111111111111111111111221121111111221412214122111
2111122211121412211221111222111111112211121121421241424241211
4221211111411111141111241112121212221112222122214122221221212
2421212241114214422121111122211121241111412122421111124122122

#42 (mean 1.579, $|D_3|$=0.238):
2222222222222222222222222222222111221114112111111114112412111
1212411124111221121211111121112112111112221111211221211212121
2111141211111122111122122121212111212111211212111111212211111
2122411124111221222122111122111242221141121212122422222124411

#43 (mean 1.579, $|D_3|$=0.215):
2224222422211212242112141412112111244112121111121111124111222
1111122211112224122111111111212121142112111121211222421422211
2211112121111221221121211112111121221111222122112111111222144
2122412112111112212121111211211422111111142111112122214121211

#44 (mean 1.592, $|D_3|$=0.215):
1111111111111112111221221122111114211122111122421112121141111
4222111122111112224421111221111212241211121222141221142221111
1122142112211112111122122411212112212211211221211121111212141
1121121111214111122221141222211212122212244114422212221121121

#45 (mean 1.587, $|D_3|$=0.217):
1211211224212211141411111111112121224112211111111112111121111
1221122222112112122241112411412222412421412112211212214214121
1111111222122121114221111212211121121141212111122222222114121
1111111122121211211111212141112411111411212221121422111111411

#46 (mean 1.579, $|D_3|$=0.215):
2222422222222222222222222121111121111211214221114411221111112
1241111111121211111211122221111121221211211112211112241221141
2111211444111114212211122221111121111112122121121112222211111
2141121412412141121142111121212112111211421121111121111212121

#47 (mean 1.579, $|D_3|$=0.229):
2242222222222222222222422244222222212211112411114411421111111
1111212112111112414121111121221212111122211212112112221112111
1114111212111221111112212122221122112121121111111111212211111
1214211211222122421112222122221212211111111111211121211214111

#48 (mean 1.579, $|D_3|$=0.215):
2422222224212211121111111112121112441141111111224221214211121
2111212222122212411121122122122121121212212124211111411112111
1214141111111111121114112112122221421211121221121222121112211
2424212212111111111111122221111222112111141121111121122121121

#49 (mean 1.592, $|D_3|$=0.215):
1111111111411211422221221144112112214212142421211212442111111
2122211211211212214212211411212111212212121122222212121112111
2211111211221111111121121241112111221111122121211121211111221
2222111121141212222114421421222111112411112222412121211111111

#50 (mean 1.592, $|D_3|$=0.216):
1221121214211112421211221121212112212122214214211111211111211
1412211111414211112221221214121221111144112111111111111212211
2111441211121111211212212211222212114121211111111121114111212
1214112421242122121221221241122111141211121121224211111221211

#51 (mean 1.579, $|D_3|$=0.229):
2222222221221242111224122211221121121112211222141221111121121
2111244112111214211121111411112211111222442111111111211222111211
2121221211111222222111221211412212221121221112112111221122211
1111121222211411141141121211111214112221211212211221111212112

#52 (mean 1.579, $|D_3|$=0.248):
2224222222222222222222222222222222222222221221112111221111112121
2111222111111121112411111222211211111112222111121121122111211
2421111111221121224212221124421124141211112211122111121111111
4111122122111212111111241122112112121144112111211211121221121

#53 (mean 1.592, $|D_3|$=0.218):
1111111111111111142121114422111211121211211421221411111121111
4121221214222222222121111212112122222214111122121211141122212
2144111142411111214221112212121112121112121212411112221112 12
1112122121122211212242222121221141112111221221122212222111211

#54 (mean 1.579, $|D_3|$=0.215):
2422222244222222244222222222224211111212221111111211111111142
2112111121222211121211111421122211411111211221221121111122111
2112421112212111111212111421211111412121211211111211111221122
1111442212212111422111121121111222111111221121114121111211421

#55 (mean 1.592, $|D_3|$=0.233):
1111111111111112114411114141214221121211122111121221112111111
2211212111112221211111122112112142111224121111222122111411112
2111121421121212411122142241124112212412111121121121111212112
2211411112422412121412111114112114242221211221141141111121121

#56 (mean 1.592, $|D_3|$=0.279):
1111111111111111111111111111111111111111111111111411112421241112
2411221111111211211421111141212242211111221212442111121111141
4421222214111211211221121421124224111142112142112124111112411
2111111222121221221412111112212144122112122114414111111211142

#57 (mean 1.579, $|D_3|$=0.215):
2222222222112141122112112211111111211141122212224211211121111
1112114211112222211212412221221411211212411111121112112 41222
2141221121121411111111121421124211111221214111212221412112 11
1221121112214221211111422221242112211211211121112121111111111

#58 (mean 1.579, $|D_3|$=0.218):
2222222222222111112121114441211141211221212111121122112 41244
1221114111121121112112111111112122212142122111122121114411 12
2211211122111242211211224122212111111111142121112421112412112
1211211221211121111221212112221122111221211121211221122221111

#59 (mean 1.579, $|D_3|$=0.238):
2222222222222222222222222222222222222222422222112211111112211
1121121221121121444112112111141212411121122121112112121212111 1
1124211142114222112111221221111211241112121141211111111121212
1111121221111112111114211112211121112111111114211211121121211

#60 (mean 1.579, $|D_3|$=0.214):
2224222222222222211221112421112111241212211142111211211141221
1111121121214112211111111122412111142121411411111221111141112
1112112122114141121114412212121112121122111242122211111122221
1112421121121111121211112114121111112211111222112212222221211

#61 (mean 1.579, $|D_3|$=0.238):
2222222222222222422111111111211411112121122211142421211111111
1211221121242114142211112222211122111111112122111111112112221
2122214212111122122211221221111211111111122142211122111112222
2121222212112111211422122121441221112112111221114211212121121 12

#62 (mean 1.583, $|D_3|$=0.230):
1111111111111111111111111111112412211111211421211224214141 1212
1112212221221411111211114222114121141114112121111212112122 41
1122114124111121111224211412121121212224412242111141212111411
1221221111122112411211211111221112221211212112122122221114122

#63 (mean 1.579, $|D_3|$=0.222):
2222222222222222222222114121411211222112211112111112241112211
1112211111212122141422111221114112112411121122111112142112211
2121142141111111212111121111411142142222211121111212212411 11
2221224111121121122111111111122121112111222122112221211122 11

#64 (mean 1.579, $|D_3|$=0.215):
2222222222222222222222111114111121412114111112121224121111 2
2111211211112142111111412221221211111441111111114112111242112
2111114122112211411222111211214111421121121122111111221212211
2212121121121111211111141124221122112122111214221112122212 11

#65 (mean 1.579, $|D_3|$=0.222):
2222122211212111111412112141111222121111112121121221221 22211
1112424112421211122111212111111122111211212222221121111212211
1121121221122122122111112211212222141121112121141221212112 11
1421214211141122122222221211124112112111141422112111111441221

#66 (mean 1.592, $|D_3|$=0.216):
1111111111111111111111111111111111111111111111112121211122111111
2211222124122421122111121112111112212224111422212121122224211

```
1241124112114411211211122242141211411112211222222121212122
2222112121142412124424211211121214212111242112422212121122
```

#67 (mean 1.592, $|D_3|$=0.215):

```
1111121121114111111222212121114114111211211111212124114122221
1222221112122141141122121222212112114112121124112111221122122
2141112141112212211121411121111212112112212241214111222111121
4121111442112121112124122112121222121112121222112221112112222
```

#68 (mean 1.579, $|D_3|$=0.216):

```
2222222222224224242411121211112124122114421211211411111122
2121121211111124421211221222221121211121112121121114121111122
1112121114122121111412111211111112212211122111111121124112122
1221211111141111111441122111111112242121121241112221111121111
```

#69 (mean 1.592, $|D_3|$=0.216):

```
1111111212241111221211114111112212112122241122212222221142111
1111142212112422121411222242422112142421112211112212112221111
1411111121212122112211111212211111221141111112422121112111111
1212411222112141111111411122421121112111412121121121111421212
```

#70 (mean 1.592, $|D_3|$=0.216):

```
2111421211224122241112241121121121211211212112111114121122222
1112211121214121221212111111224112121121212111112211121422424
1122211212112121211422112221112112114111422211122241111111111
1241221211141211111221212211211121442111111442241211111111112
```

#71 (mean 1.592, $|D_3|$=0.214):

```
1111111111111111122112141214211121221122111141221442211222222
1144144124242112221411121121212111111211142122211122212122111
1222242112212111212222124111114112112211121121111212111112222
2112122112142121211112111221121241211211122141212111211111112
```

#72 (mean 1.579, $|D_3|$=0.214):

```
2422422222122111114111211112411211211212122112211111221212121
2244111142211221112241111112121211112112121142121211412421111
2121112112221122214122411121111121221422121211212122111122121
1222122121212112111112111214112111121112241111111121141111114
```

#73 (mean 1.587, $|D_3|$=0.221):

```
1211411111224111211211242211212122114112221111211111124214211
1222211111111221211121214121114411122121111212111112212121211
1122111111124124212112221221111141211241111111112111211212411
2111222422141111224112112111212221412121222411112211114112411
```

#74 (mean 1.592, $|D_3|$=0.215):

```
1111111111111111111212211212142222211411112212121121211221111
1112212211412421122221111221221212112122241211222211422122121
2122212211121211111122242122111111112121212121111112114244142
2214112111241112114111121111111122111141224221222211411211124
```

#75 (mean 1.579, $|D_3|$=0.215):

```
2211241114212121212121112222111111124212112121211211412212111
2121114121141111212111121141121241211241111211212212212244121
2211112222111121221221111112121212222211411121114222211112111
1142112211211212122121112122124124112121111212211211111141211
```

#76 (mean 1.579, $|D_3|$=0.225):

```
2222242111221411122111112421111111211121121241111112141111122
4211111211112121211111224121112142121211121111111122211221222
2111221212212111411211124221242111241122112111111111121124142
2111412121121111121111212221214142211112112211144122144211121
```

#77 (mean 1.583, $|D_3|$=0.241):

```
1111111122221112111112112111122212114141222421422222212421422
2211221141242212121221121111241211121212111211211111212222211
1211221121211422221114222211112411112112111221111112212211222
1212212211242121112121122122121112212211122121111221212121122
```

#78 (mean 1.592, $|D_3|$=0.215):

```
1111111111121121112122211442211222212222221111121122121114122
1122211222121142414111122212111121114111212441111112112222111
2212222121121211121122214412211111111412122221111121222212222
1111224121112211121111211211114221222142141111221111114411211
```

#79 (mean 1.587, $|D_3|$=0.221):

```
1412112211111114212212121112121122221112111111111141111111124
1121141422212211111211221421222212111211221111111144141242222
1111112411214111211122111111242114121111421112212111221121122
1241121411122211122121221121112411411121212212211211112222422
```

#80 (mean 1.587, $|D_3|$=0.217):

```
1111121442111122111412121121112112221111221212111212211111211
2112211111214142122121114221422121112112111211221121211221422
1211122122121222211412222121211121111112212411111142111222422
1221122422211124111421212121121124242121211122111211111211122
```

#81 (mean 1.592, $|D_3|$=0.225):
111111111111111111111241411112222112141212121111412114211421
112212111111121121121241211111224241111111221142112111212141
222211211121142211122241112422211111114122112122124122211112
111112121112221211224114211124221242122214211111121412211121

#82 (mean 1.583, $|D_3|$=0.214):
212112124241211112141121211212211211212112124221111111121214
224111211121124412221122122111222112121122141142214112424421
124211221112111112112224111121121212111221112111112112212121
111111111221221112141122111211112221111122112241211221221111

#83 (mean 1.592, $|D_3|$=0.215):
112212111124212112211211111211211111411221212122211111111122
221111224111222212221222121111212222112112111422421441221212
241112221221211121414122442221112241111211212112211112111421
411112121121121141411111221212214112112121111112221112121111

#84 (mean 1.592, $|D_3|$=0.218):
122111111121141121122212142221142212112221111212122121211211
121211112112122421411111141121242211112122221111211144112242
121111111112212112212222122114412211111212221221114111111214
211121214221142111212111211121141121111222121122222221112221

#85 (mean 1.579, $|D_3|$=0.222):
212122441122222114222122121211211111211112111414114121111122
122121111121121221211211211212211211412221211211121122221211
222124212211211121121124121214111122124111122112212111112212
441221221212111122121224111212121221212221112114121212111111

#86 (mean 1.579, $|D_3|$=0.238):
222222222242222422222222222221122111121224121221112124211221 4
112111111111111211221421111112211121111112121121412221111142
211221111122111121112121112141221212121121111221212212211 2112
212122221111411424212211212112111112222212121121114111112112

#87 (mean 1.579, $|D_3|$=0.229):
422222222222222224222422221422221112221212111211211122111412
111221112111111422111221111112112411211112212241111111121111
112122211111111111111141111224111211221211111112211211242222
221122221111112111211241421121221122122242211112121212211221

#88 (mean 1.587, $|D_3|$=0.227):
242141212121121212141211141112221212111112122112121212122222
121112221111211111121241211121111221221111111121111211212122121
121242122412212111212122211111122121222211212421212121221122
112224211112222121244222111112141412111111211422111111124122

#89 (mean 1.579, $|D_3|$=0.218):
422222222211411121122111212121122111111121211111111122122121
121122211212122122211112412112111411211421111211212111121114
241221111122111411111224141121211121111242221222222212122242
211221111211211221111112221121111212241114214111241222211 12

#90 (mean 1.592, $|D_3|$=0.216):
111111112112121114212212121112122112211121211111214221114211 1
211122111124412111222214212111111112121121121142211112421112
111121212212222121224142112211121212121212121112111411421211
111121141112212121111222221221411411112244142112111241224111

#91 (mean 1.579, $|D_3|$=0.216):
422222422221214111212112121214111211114111212111211111211112
112112121221241121111214122121211111111111121122241112111 22212
124412214142114412121121111112412211121111122221221212111211
221221111112412211111111211122112111112212114121114214121222

#92 (mean 1.583, $|D_3|$=0.225):
111111212222111221221141121122112242122221111124221211112112
124111222411211241121122112111111112122412242211111111112212
211121112212121112112111122121112121222422222222211122121212
112112112221241242421141211112111111221111121242111241211221

#93 (mean 1.592, $|D_3|$=0.220):
111111111111111111111122124212111111414121222121111211221211 1
112214111121221111112221241211211112412112121212212122112122
442421122111421124112121111221121112112114111244114111124122
221112111211222122121211424212242121411212111411122112122111

#94 (mean 1.579, $|D_3|$=0.214):
442222222222222224224121111111111111211122111121124142112211211
112112122121224212122411211112211122124212112111411121111211
141242112111112122111122241141211111241221111111111142111112
222111221111121211121421112122212211112112111121412212112211

#95 (mean 1.579, $|D_3|$=0.238):
222222222222224222222224222222222222222222211211111111112212121
241111121111121121221112411211211111214212112112121112111221

1112412222111114111221221111114222412112111121211211111111112
2111122122211111411122122211112422224111111111211122224122111111

#96 (mean 1.579, $|D_3|$=0.238):
2222222242222222222222121122421111211111212112124121211122122
2211222212111121114122111111112141122111221121212211111221112
1211141221141111221112112211111222112124141211212124221211141
1222121111111211241221111122221211211111111221121112421122121

#97 (mean 1.592, $|D_3|$=0.233):
1111111111111111111121141422211112112211122221111121111142412
2112114214211142121212241114221241112124211142211222122111112
1122112111111121121212221112111141212112222141214111111211221
1214111112221411111122211221111411211412112411212244111112411

#98 (mean 1.579, $|D_3|$=0.229):
2222222222222222222222211114121111112211122122211111112121221
1212111422112111221111412221221111221112111111211121211111122
1112122414111111114122222212111112111111212112112222114111124
1241212224112112111124224122121142112111221112241121111212212

#99 (mean 1.592, $|D_3|$=0.218):
1111111111111111111141122114142422421111124124111122112111112212
1221111414112111221111241221211411211111221121244222221224221
1222121222121211111112112222242141222112211122211221112122212
1221112111121222212122221112211112111222214111111211212122111

#100 (mean 1.579, $|D_3|$=0.215):
2222222111144112112422112111412122112221214111212111111121221
2141122121222122212221222211121214211221221112441141121111111
1144111141121114114111111421122112211211211121212121212122111
2121121111111212122121111112221122212212211112111211214112111

#101 (mean 1.579, $|D_3|$=0.218):
2222222222222111112122121111211211142121121211111221141241211
1211141111121121221111111212124112222112221121211111211122111
1422221121121121114111211121212421222221111424222122114212221
2224111211114121412121211112111121111121212112112111424122111

#102 (mean 1.579, $|D_3|$=0.218):
2222222222222222222222222242222222222212121122421122212241211
2121211112112121112212211411211111411242111111211112111141111
1211221111121111112111114211211142111111111122411124442112111
1122111111112111211214111112111124241122111111221121211121121

#103 (mean 1.579, $|D_3|$=0.229):
2222222222121111112111211122411222221121211122122111122111111122
1221111212421121121211121241212121121222442121112121442211
2112121211111112422121121121211242211112211111222111121211112
1214121111211212122112221112412221241211142111111121241121112

#104 (mean 1.587, $|D_3|$=0.217):
1211422221212121221211142111221111221221122214121121212222221
1111214112112112211411111222121111221111121111221222222
2114212112111111211211112141114214112122211122111121421241221
2111112141222121141212112221121111142211411221212212242111141

#105 (mean 1.592, $|D_3|$=0.229):
1111111111111111111111111111111121212211111212211221121141121
1121211111111211211122212224211121421124212221121222422222
1124212121111121222112244111222224212222112111222124221222241
2221122411211212411122121212124211222111211211121212411222121

#106 (mean 1.583, $|D_3|$=0.232):
1111112114222212212121112411111211222241142211221221114221111
2111222212111221122212221121211112222121112121212124222111222
2121221111121122111112212111122211142242112142211221124121142
1221411141212112112221111112111112242111221211111221112211112

#107 (mean 1.579, $|D_3|$=0.218):
2422222222221221111111111222141212222111111221122111111111111
1212111111114212112211111142222411212111121111111111121212221
2221212221111211121211422111212121211412142112111221212212211
1121111111441221221111422221214122221111242221141222221111424

#108 (mean 1.579, $|D_3|$=0.214):
2422222222222222121221211221111212112212211211111222421441412
1112111111211111411211112221111111221421111221111111142211122
1111111222222221222114141111414211121211111111211212141121
1111221414411112221111121211121111111242112222111111211211141

#109 (mean 1.592, $|D_3|$=0.220):
1111111111111122421242142141122111214212222121122111242111214
2111212111112214114142121112124211111111111221221111242142121
1121422122221111211111112412422111111112121121111212211211212
4121111112211112111211111112121114242221141224212222111121222

#110 (mean 1.592, $|D_3|$=0.220):
1111111111222142112122212212111111411214211124124112111142121
2111112111111111124112121211111222222124112111114222221221122
2422222111111211421211111111211121122442111121111112121124211
1122122111121111112141112214211421221221242114212141121112 14

#111 (mean 1.579, $|D_3|$=0.229):
2222222121221211211112121121112111111121142122212211111141212
1241121212212111111221212211241111122212211121121111211222111
1224141111111122112444121221212111112112242122221121221122 21
1121212211122112121121212124112121122111212421112112214 22421

#112 (mean 1.583, $|D_3|$=0.218):
1111111111121211121141221221412211122211111121114111111122111
2411214212111121212124114221124221111211122221441412142221 21
1121211212221112221121112112111111112122111111211211111121121
2124421114111222121221122211222111124424221112111442212 11111

#113 (mean 1.579, $|D_3|$=0.248):
2222422222222222222211212221112222111211111422111212121112 41
1111122121212122421111211222111221121122121211112112211111111
1122221122111121121222112111114221142121221121414111122111 12
2124112112121111211221121122141222211214222121214222111111 12

#114 (mean 1.579, $|D_3|$=0.229):
2222222222222222221111112112112111121212121241111224111122 22
1122222122111112111112112121114112221141241111112221124112 11
2112142121111111111114412112221221111212114212411212111221222
1422121211222212122211212112121211141112112112112111112214211

#115 (mean 1.592, $|D_3|$=0.220):
1111111111111111112112112111422222122122222422122112111114111
2222121211121111111111441114221212121411124221412421121121 12
1212114111212212221111112412144221112221221211141211212412 11
1222114411121121211121112111111211114211112224114222112111 11

#116 (mean 1.592, $|D_3|$=0.233):
1111111111111111111111111111111111111111111211211211421222 1244
1121114221241111212412221211422221212421221121212221112222 12
2214144121141111111212124211412142141212421221111112111122 22
1121111111121141211211124111112242221411212221241241111211 11

#117 (mean 1.579, $|D_3|$=0.222):
2222222211222211114111112221421122121242211221111111222412 21
2112112112112221112111121211112111121111222122421111112121 1422
1111111211112122224111111211211221111121112142111112211212 22
4122221122121411111214122222111112121141242112412221141122 21

#118 (mean 1.587, $|D_3|$=0.214):
1241222422221411121211111121212112122111121211121111121422111
2112211111211111112214111122211112221211111211121111124221 2111
1112211242111221111114112111212211112142112242211122222122 4144
1242212222111111421211112111112424112122222111111211212212 1142

#119 (mean 1.579, $|D_3|$=0.218):
2222422222222222422222222222222222242112112112141111111112 21142
1112121112111212122212211211122121112211121121111111121124111
1112111122111142111111121111111241421224221411112222121111122
1411211221111111121121211412112211111121121111111414222412 11

#120 (mean 1.579, $|D_3|$=0.214):
2222224222242222221224112121221111211111111121211211212121 11
2122111211211121121111112114122112211121122111222414111121 11
1211221212111221141211212114122111114111122121111112114224 44
4112111111212112222411412212412411121211212111111121121211221